PHYS 143 – Mechanics and Thermal Physics

The First Law of Thermodynamics

How does a system exchange energy as heat and as work, and how does the first law connect these to its internal energy?

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By the end of this week, you should be able to:

1. Describe a thermodynamic system, state and process, and calculate the work done by a gas from a pV diagram, including its sign and its dependence on the path.
2. Apply the first law, $\Delta U = Q - W$, to a process and to a cycle, using an energy table of Q, W and ΔU.
3. Calculate Q, W and ΔU for the isochoric, isobaric, isothermal and adiabatic processes of an ideal gas, using C_V, C_p and γ.
4. Analyse a multi-step process or cycle in an unfamiliar problem, choosing the physics without being told.

The detailed outcomes are in the checklist at the end. **Reading:** Young and Freedman, Ch. 19 (§19.1 to §19.8) and §18.4.

Labels: **CORE** you should be able to do this; **PRACTICE** use it to develop the skill; **REFERENCE** keep it for later use.

Two contacts:

A (Tuesday) REPRESENT, then QUANTIFY
How does a gas exchange energy with its surroundings, and what does the first law say?

B (Wednesday) QUANTIFY, then APPLY
What does the first law give for each process, why do C_p and C_V differ, and how do we treat adiabatic processes and cycles?

Practice: check → example → independent → application

Not marked. Answer individually, then we review.

1. A constant force of $20\,\mathrm{N}$ moves an object $0.30\,\mathrm{m}$ in the direction of the force. Find the work done by the force.
2. Find the area under a horizontal line at $y = 3.0$ from $x = 1.0$ to $x = 5.0$, and under a straight line from $(1.0,\, 3.0)$ to $(5.0,\, 1.0)$.
3. Evaluate $\displaystyle\int_{2.0}^{6.0} \frac{\mathrm{d}V}{V}$.
4. One mole of a monatomic ideal gas is heated from $300\,\mathrm{K}$ to $400\,\mathrm{K}$. Find the change in its total translational kinetic energy.
5. A gas at $1.0 \times 10^5\,\mathrm{Pa}$ has its volume doubled at constant temperature. What is its new pressure?
6. Convert $4.0\,\mathrm{L}$ to cubic metres, and find $p\,\Delta V$ in joules for $p = 2.0 \times 10^5\,\mathrm{Pa}$ and $\Delta V = 4.0\,\mathrm{L}$.

1. A force of $20\,\mathrm{N}$ moves an object $0.30\,\mathrm{m}$ along the force: $W = Fd = (20)(0.30) = 6.0\,\mathrm{J}$.
2. Area from $x = 1.0$ to 5.0 under $y = 3.0$: $3.0 \times 4.0 = 12$. Under the line from $(1.0,\ 3.0)$ to $(5.0,\ 1.0)$: average height 2.0, so $2.0 \times 4.0 = 8.0$.
3. $\displaystyle\int_{2.0}^{6.0} \frac{\mathrm{d}V}{V} = \ln(6.0/2.0) = \ln 3 = 1.10.$
4. One mole of a monatomic ideal gas heated from $300\,\mathrm{K}$ to $400\,\mathrm{K}$: $\Delta K_{\text{tot}} = \frac{3}{2}nR\,\Delta T = \frac{3}{2}(1)(8.314)(100) = 1.25 \times 10^3\,\mathrm{J}$ (Lecture 10).
5. A gas at $1.0 \times 10^5\,\mathrm{Pa}$ with its volume doubled at constant temperature: p halves, $0.50 \times 10^5\,\mathrm{Pa}$ (Boyle).
6. $4.0\,\mathrm{L} = 4.0 \times 10^{-3}\,\mathrm{m}^3$; with $p = 2.0 \times 10^5\,\mathrm{Pa}$, $p\Delta V = (2.0 \times 10^5)(4.0 \times 10^{-3}) = 800\,\mathrm{J}$.

Work and energy (Lectures 4 and 5), the gas model with $K_{\text{tot}} = \frac{3}{2}nRT$ (Lecture 10) and $Q = mc\Delta T$ (Lecture 9) are reused. That is the toolbox for this lecture.

But what if... a gas ends in the same state whether it is warmed by a flame or squeezed by a piston? Lecture 10 left us asking how much work a gas does when its volume changes, and where the energy comes from. A puzzle shows why the state alone cannot tell us.

Contact A: Represent, then Quantify

How does a gas exchange energy with its surroundings, and what is the same on every path when heat and work both depend on the path?

Poll

The air in a closed bicycle pump can be warmed in two ways: hold the barrel over a flame, or push the handle in quickly. Suppose both end at the *same* pressure, volume and temperature. **Can anyone, by measuring the final state, say which method was used?**

Predict first

Sketch the pump twice, once with a flame under it and once with a hand on the handle. What enters the air in each case? What do p, V and T record?

No. The state (p, V, T) records *where the gas is*, not *how it got there*. In the first case energy crossed the wall as **heat**; in the second as **work**. Heat and work describe a *process*; they are not properties of the gas. Today we count both, and relate them to the energy the gas holds.

But what if... we want to count the energy that crossed in each way? First say exactly what "the gas" is: the **system**, its **boundary** and **surroundings**.

- A **thermodynamic system** is any collection of objects that is convenient to regard as a unit, and that may have the potential to exchange energy with its **surroundings**.
- The **boundary** separates the system from its surroundings. In this lecture the system is usually a quantity of gas in a cylinder with a movable piston.

Choose the system first

Define clearly, at the start, exactly what is and is not included in the system. Only then can the energy transfers into and out of it be described unambiguously.

But what if... energy and matter can cross the boundary in different ways? The kind of boundary decides whether a system is **open**, **closed** or **isolated**.

Open, closed and isolated systems

- **Open** system: matter and energy cross the boundary (a cup of hot tea).
- **Closed** system: energy crosses, matter does not (a sealed bottle; the gas in a pump).
- **Isolated** system: one that does no work on its surroundings and has no heat flow to or from them (an ideal vacuum flask).

Walls (Lecture 9)

A **diathermal** wall lets heat through; an **adiabatic** wall does not. A piston lets the system exchange **work** with the surroundings.

But what if... a system is not described by a single number, and a gas changes as we heat or squeeze it? We need **states**, **processes** and **paths**.

- A **state** is specified by the state coordinates p, V, T (and the amount n; Lecture 10). It is defined only when the system is in **equilibrium**: thermal (one temperature), mechanical (uniform pressure) and chemical (no reactions).
- A **thermodynamic process** is a process in which there are changes in the state of a thermodynamic system, from an initial state to a final state. The intermediate states between them form the **path**.

Physical meaning

An explosion or a sudden rush of gas has no single p and T in the middle of the process, so it has no line on a pV diagram. Only a **quasi-static** process (defined next) can be drawn as a path. In this lecture every path is quasi-static, unless we say otherwise.

But what if... we keep saying "work", "path" and "quasi-static" as if everyone meant the same thing? We need exact definitions: the **vocabulary of thermodynamic work**.

Defining the vocabulary of thermodynamic work

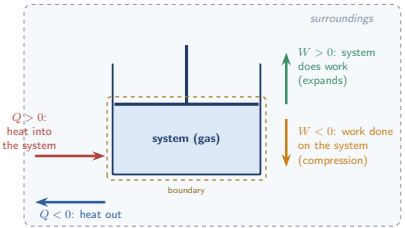
CORE

Quantity	What it is	What it measures	SI unit	Representation
Work done by a gas W	the work done by the system on its surroundings in a volume change: $\mathrm{d}W = p\,\mathrm{d}V$, so $W = \int_{V_1}^{V_2} p\,\mathrm{d}V$; positive when the system expands	how much energy leaves the system as work (negative: enters)	J	area under the path on a pV diagram
Path	the series of intermediate states through which the system passes from its initial state to its final state	how the change is carried out	—	a curve on the pV diagram (when all the states are equilibrium states)
Quasi-static process	a process in which all the intermediate states are equilibrium states, so it is slow	whether the path can be drawn at all	—	a continuous line

Interpret: the work done by a gas answers *how much energy did the gas give to the surroundings by pushing the boundary?*

But what if... we know what W means, but not which way the energy goes when W or Q is positive or negative? We need a **sign convention**.

Energy crossing the boundary: signs for Q and W



Rule: sign convention

$Q > 0$: heat flows **into** the system (energy enters); $Q < 0$: heat flows out.
$W > 0$: work done **by** the system against its surroundings, as an expanding gas does (energy leaves).
$W < 0$: work done *on* the system by its surroundings, as in compression (energy enters).

Explain. Q and W are energy *in transit* across the boundary, measured in joules; neither is stored in the gas. Energy entering as heat ($Q > 0$) raises the energy of the system; energy leaving as work ($W > 0$) lowers it.

But what if... the sign of W here is not the sign we used for work in mechanics? One reversed sign spoils every later step, so we state the difference plainly.

Two sign rules for work

In mechanics, W was the work done by the forces acting on an object. In thermodynamics it is usually more convenient to call W the work done *by* the system, so that when a system expands, the pressure, the volume change and the work are all positive. The two rules have opposite signs.

	expansion	**compression**
work done *by* the gas, W (this lecture)	$W > 0$	$W < 0$
work done *on* the gas (the mechanics sign)	negative	positive

But what if... every problem needs the same two questions, in the same order (what is the system? which way does the energy go?)? We make them a habit: **Toolkit 1**.

- **TOOLKIT 1 – System and sign convention**

1. SYSTEM: name it (the gas), draw the boundary, and say whether it is closed (fixed amount of gas).
2. ENERGY IN OR OUT? For heat: does energy enter ($Q > 0$) or leave ($Q < 0$)? For work: does the gas push the surroundings ($W > 0$, expansion) or do they push the gas ($W < 0$, compression)?
3. UNITS: Q and W in joules; $1\,\mathrm{Pa\,m^3} = 1\,\mathrm{J}$; $1\,\mathrm{cal} = 4.186\,\mathrm{J}$; T in kelvin.
WATCH OUT: the mechanics sign for W; using a temperature interval ΔT where T is needed (and the reverse).
CHECK: expansion gives $W > 0$, compression gives $W < 0$; heating the gas gives $Q > 0$.

But what if... we can say which way energy crosses; but can we tell, without any integral, whether a gas does work in simple cases? **Decide first.**

Concept check: work by a gas before the integral

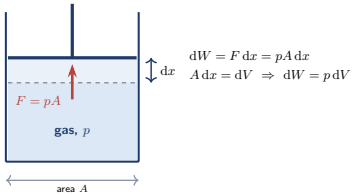
Decide first

1. Does a gas do work when it expands into a vacuum?
2. Does a gas do work if its volume stays fixed, however high the pressure?
3. Can a gas do work while its temperature stays constant?
4. Is the work done by the gas the same as the work done on the gas?

Answers

1. **No.** There is nothing to push: $p_{\text{ext}} = 0$.
2. **No.** No volume change, no work, whatever p is.
3. **Yes.** An isothermal expansion: heat flows in to pay for it.
4. **They have opposite signs:** $W_{\text{by}} = -W_{\text{on}}$.

But what if... "no volume change, no work" says nothing about *how much*? We need the work for a given volume change, starting from the force on the piston.



Rule: work done by a gas

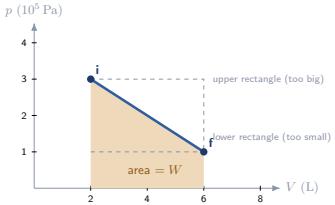
$$W = \int_{V_1}^{V_2} p\,\mathrm{d}V$$

Constant pressure: $W = p\,(V_2 - V_1)$.
Constant volume: $W = 0$.

Physical meaning

The force on the piston is $F = pA$ (pressure times area). Moving the piston by $\mathrm{d}x$ gives $\mathrm{d}W = F\,\mathrm{d}x = p\,\mathrm{d}V$. Expansion ($\mathrm{d}V > 0$) gives $W > 0$; compression gives $W < 0$. No volume change, no work.

But what if... the pressure changes as the volume changes? Then $p\,\Delta V$ with one pressure fails, and the work becomes the **area under the path**.



Rule: work as an area

The work done by the gas is the area under the path on a pV diagram. Moving to larger V gives $W > 0$; moving to smaller V gives $W < 0$.

Common error

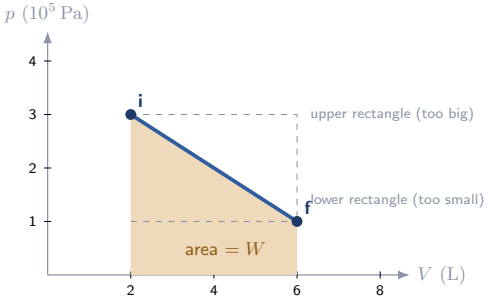
Using $p\,\Delta V$ with the final (or the initial) pressure on a sloping path. p changes along the path.

TOOLKIT 2 – Work from a pV diagram

1. **SKETCH** the path with an arrow from the initial to the final state; mark the volumes.
2. **SPLIT INTO LEGS:** horizontal $W = p\Delta V$; vertical $W = 0$; straight sloping line: average pressure times ΔV (a trapezoid); curve: integrate (isothermal: next frames).
3. **SIGN:** moving right (V increases) is positive; moving left is negative.
4. **UNITS:** p in pascals and V in m^3 (litres must be converted).
5. **CYCLE:** the net work is the enclosed area; clockwise is positive, counter-clockwise is negative.

WATCH OUT: V left in litres; forgetting the sign of a leg that goes to smaller V.
CHECK: the answer lies between the areas of the rectangles that bound the path.

But what if... a routine is not yet a result? We test it on a real path: **Example 1**.



A gas expands from an initial state i ($V = 2.0\,\mathrm{L}$, $p = 3.0 \times 10^5\,\mathrm{Pa}$) to a final state f ($V = 6.0\,\mathrm{L}$, $p = 1.0 \times 10^5\,\mathrm{Pa}$) along a straight line on the pV diagram. Find the work done by the gas.

Predict first

Is W positive or negative? It must lie between the two rectangles shown. Between which two values in joules?

Example 1

A gas expands from i ($V = 2.0\,\mathrm{L}$, $p = 3.0 \times 10^5\,\mathrm{Pa}$) to f ($V = 6.0\,\mathrm{L}$, $p = 1.0 \times 10^5\,\mathrm{Pa}$) along a straight line on the pV diagram. Find the work done by the gas.

Set up: $\Delta V = (6.0 - 2.0) \times 10^{-3} = 4.0 \times 10^{-3}\,\mathrm{m}^3$. The bounding rectangles: $p_f \Delta V = (1.0 \times 10^5)(4.0 \times 10^{-3}) = 400\,\mathrm{J}$ and $p_i \Delta V = 1200\,\mathrm{J}$.

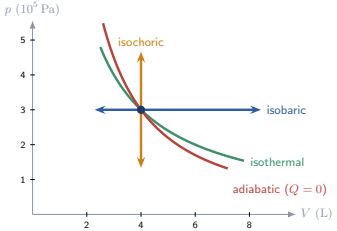
Execute: the area is a trapezoid, with average height $\frac{1}{2}(3.0 + 1.0) \times 10^5 = 2.0 \times 10^5\,\mathrm{Pa}$:

$$W = p_{\mathrm{av}}\,\Delta V = (2.0 \times 10^5)(4.0 \times 10^{-3}) = 800\,\mathrm{J}.$$

Evaluate: $W > 0$ for an expansion, and $400 < 800 < 1200\,\mathrm{J}$, exactly halfway between the rectangles because the path is straight. The units work: $\mathrm{Pa\,m^3} = \mathrm{J}$. ✓

But what if... the path is not one straight line, but flat and steep pieces or a hyperbola? The paths that occur most often have names and simple rules: the **four special processes**.

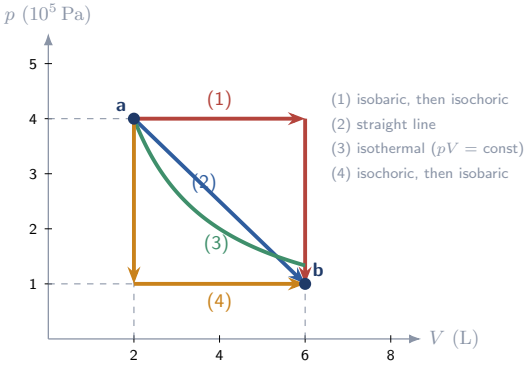
The four special processes on a pV diagram



Process	Fixed	Path
isochoric	V	vertical
isobaric	p	horizontal
isothermal	T	hyperbola
adiabatic	$Q = 0$	steeper curve

The first three are named for the quantity held fixed; in an adiabatic process no heat is transferred into or out of the system, $Q = 0$. On a pV diagram the paths are called an isochor, an isobar, an isotherm and an adiabat. The adiabat falls faster than the isotherm (Contact B).

But what if... two states can be joined by any of these paths, or by combinations of them? Does the work depend on which path we take?



A gas goes from a $(2.0\,\mathrm{L},\ 4.0\times10^5\,\mathrm{Pa})$ to b $(6.0\,\mathrm{L},\ 1.0\times10^5\,\mathrm{Pa})$ by four different paths. Same start, same end.

Predict first

Which path gives the greatest work, and which the least? Is the work the same for all four?

Question restated

A gas goes from a $(2.0\,\mathrm{L},\ 4.0\times10^{5}\,\mathrm{Pa})$ to b $(6.0\,\mathrm{L},\ 1.0\times10^{5}\,\mathrm{Pa})$ by four paths. Which gives the greatest work, which the least, and is W the same for all four?

$\Delta V = 4.0\times10^{-3}\,\mathrm{m}^3$ on every path.

Path (1) (isobaric at $4.0\times10^{5}\,\mathrm{Pa}$, then vertical):
$W_1 = p_a\Delta V = (4.0\times10^{5})(4.0\times10^{-3}) = \mathbf{1600}\,\mathrm{J}$.

Path (2) (straight line): $W_2 = \frac{1}{2}(4.0+1.0)\times10^{5}\times4.0\times10^{-3} = \mathbf{1000}\,\mathrm{J}$.

Path (4) (vertical drop at $2.0\,\mathrm{L}$, then isobaric at $1.0\times10^{5}\,\mathrm{Pa}$): $W_4 = p_b\Delta V = \mathbf{400}\,\mathrm{J}$.

Path (3) is a curve. Its area needs an integral, which we do next. For the ideal gas the curve is an isotherm, because $p_aV_a = p_bV_b = 800\,\mathrm{J}$.

But what if... the three works already differ, so W is *not* fixed by the end states, and path (3) is a curve whose area needs an integral? That is the **isothermal work**.

Rule: work in an isothermal process

$$W = \int_{V_1}^{V_2} \frac{nRT}{V}\,\mathrm{d}V = nRT\ln\frac{V_2}{V_1} = p_1V_1\ln\frac{V_2}{V_1}$$

Ideal gas at constant T: $p = nRT/V$, with T taken out of the integral. Since $p_1V_1 = p_2V_2$, equivalently $W = nRT\ln(p_1/p_2)$.

Path (3): $p_aV_a = 800\,\mathrm{J}$, $V_2/V_1 = 3$:
$W_3 = 800\ln 3 = 800(1.0986) = \mathbf{879}\,\mathrm{J}$.

Physical meaning

Expansion ($V_2 > V_1$) gives $\ln > 0$, so $W > 0$; compression gives $W < 0$. Constant temperature does *not* mean zero work.

Check: $879\,\mathrm{J}$ lies between $1000\,\mathrm{J}$ (straight line, above) and $400\,\mathrm{J}$ (lowest path), as the sketch requires.

Same end points, four different works

Question restated

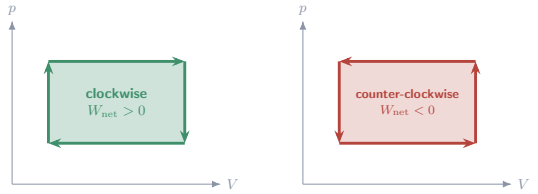
The gas goes from a $(2.0\,\mathrm{L},\ 4.0\times10^{5}\,\mathrm{Pa})$ to b $(6.0\,\mathrm{L},\ 1.0\times10^{5}\,\mathrm{Pa})$ by four paths: the greatest work is on path (1), the least on path (4), and W is not the same.

Path	Description	W (J)
(1)	isobaric, then isochoric	1600
(2)	straight line	1000
(3)	isothermal	879
(4)	isochoric, then isobaric	400

Rule: work is not a state property

The work depends on the path, not only on the end states. We write W, never "ΔW".

But what if... the gas returns from b to a by a different path? The start and end states coincide, yet the net work is not zero: a **cyclic process**.



Rule: work in a cycle

A **cyclic process** is one that eventually returns the system to its initial state. The net work is the **area enclosed** by the path: positive if the path runs clockwise, negative if counter-clockwise.

Explain. Going round the loop, the expansion leg (upper path) gives positive work, the area under it; the compression leg (lower path) gives negative work, the area under that. What is left is the area between the two paths. Going out and back along the *same* path leaves no area, so no net work.

But what if... you have never found the work around a whole loop? Try a cycle yourself.

Independent problem

A gas goes round the cycle a→b→c→a. a: $2.0\,\mathrm{L}$, $3.0\times10^5\,\mathrm{Pa}$. a→b: constant pressure to $5.0\,\mathrm{L}$. b→c: constant volume to $1.0\times10^5\,\mathrm{Pa}$. c→a: straight line back to a. Find the work for each leg and the net work.

Predict first

Sketch the cycle. Does the path run clockwise or counter-clockwise? What sign must the net work have?

a→b: $W = p\Delta V = (3.0\times10^5)(3.0\times10^{-3}) = 900\,\mathrm{J}$. b→c: $W = 0$. c→a: compression, $W = \frac{1}{2}(1.0+3.0)\times10^5\times(-3.0\times10^{-3}) = -600\,\mathrm{J}$.
Net: $900 + 0 - 600 = \mathbf{300}\,\mathrm{J}$. **Evaluate:** clockwise gives $W > 0$; the enclosed triangle has area $\frac{1}{2}(3.0\times10^{-3})(2.0\times10^5) = 300\,\mathrm{J}$. ✓

But what if... the gas retraces exactly the same path back? Is the net work then still the enclosed area? **Decide first.**

Choose, then explain

A gas expands from V_1 to V_2 along a path on a pV diagram, and is then compressed back to the original state along exactly the same path. What is the net work done by the gas?

A. positive B. negative C. zero D. it depends on the path

Predict first

Sketch the path with arrows both ways. What happens to the sign of $\mathrm{d}V$ on the return trip?

C. On the return trip $\mathrm{d}V < 0$ at each pressure the gas had before, so the integral changes sign: $W_{\text{back}} = -W_{\text{forward}}$ and the enclosed area is zero. The result holds for any path, which is why it does not depend on it (D is wrong).

But what if... these signs, areas and paths are easy to mix up? We collect the usual slips.

Common mistakes and the correct habit

Mistake	Instead
Using the mechanics sign for W (work done on).	Here W is the work done *by* the system.
Using $p\,\Delta V$ with the wrong pressure on a sloping path.	Use the area under the path.
Leaving V in litres.	Convert to m^3: then $\mathrm{Pa\,m^3} = \mathrm{J}$.
Writing ΔW.	Work depends on the path; write W.
Saying a free expansion does work.	There is nothing to push: $W = 0$.

But what if... we can now find W for any path and for a cycle, yet work and heat both depend on the path? Is *anything* the same on every path from a to b?

Poll

For the four paths from a $(2.0\,\mathrm{L},\ 4.0\times10^{5}\,\mathrm{Pa})$ to b $(6.0\,\mathrm{L},\ 1.0\times10^{5}\,\mathrm{Pa})$, the work done by the gas was 1600, 1000, 879 and $400\,\mathrm{J}$. The heat that flowed into the gas was also measured on each path. Do the heat inputs come out the same, or different? And is *anything* the same on all four paths?

Predict first

The gas starts and finishes in the same two states every time. Which quantity can depend only on those two states? Which cannot?

The heat inputs are different, just as the works are. But the *difference* (heat in, minus work out) is the same on every path. It is the change in a property of the gas called the **internal energy**. We name it first, then see the law.

But what if... (heat in minus work out) is the same on every path, but a change of *what*? The **internal energy**.

- The **internal energy** U of a system is the sum of the kinetic energies of all its constituent particles, plus the sum of all the potential energies of interaction among these particles. In a change of state, $\Delta U = U_2 - U_1$.
- Kinetic: translation, rotation, vibration. Potential: forces between molecules (and chemical energy).
- A system in a specific state has a unique internal energy that depends only on that state: U is a **state function**, so ΔU does not depend on the path.

Common errors

(1) "A body contains heat." It contains *internal energy*; heat and work are energy *in transit*.
(2) Lifting a glass of water raises its gravitational potential energy in the Earth's field, but not its internal energy: that energy arises from the interaction between the glass and the Earth, not among the water molecules.

But what if... heat and work are not stored in the gas, but U is? Test these claims before we build on them.

Decide first

1. Can the temperature of a gas rise without any heat entering it?
2. Can heat flow into a gas without its temperature rising?
3. Does a gas "contain" heat or work?

Answers

1. **Yes.** Compress it quickly: work is done on it.
2. **Yes.** An isothermal expansion: the heat becomes work.
3. **No.** It contains internal energy. Heat and work are energy in transit.

But what if... both ideas fit into one equation linking the heat added, the work done by the system and the change in U? That is the **first law**.

Rule: the first law

$$\Delta U = Q - W \qquad \text{or, for a small step,} \qquad \mathrm{d}U = \mathrm{d}Q - \mathrm{d}W = \mathrm{d}Q - p\,\mathrm{d}V$$

The change in internal energy equals the heat added to the system minus the work done by the system.

Physical meaning

This generalizes the conservation of energy (Lecture 5) to include heat. Equivalently $Q = \Delta U + W$: part of the heat stays, the rest leaves as work. Q and W depend on the path; ΔU does not.

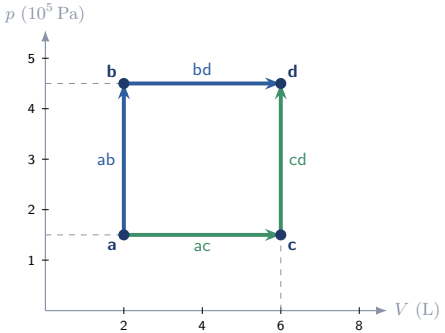
Event	Sign	ΔU
heat in, no work	$Q > 0$	up
expands, no heat	$W > 0$	down
compressed, no heat	$W < 0$	up
heat out, no work	$Q < 0$	down

TOOLKIT 3 – The energy ledger

1. ROWS: one row for each step of the process (or for each path); columns Q, W, ΔU.
2. FILL IN what is known; on every row $\Delta U = Q - W$.
3. STATE FUNCTION: ΔU between two states is the same on every path. Find it on a path where it is easy, then use it on the other.
4. TOTALS: add each column. The first law holds for the totals; for a cycle, the total ΔU is zero.

WATCH OUT: a wrong sign in W; adding a row twice; using Q where ΔU is meant.
CHECK: $\sum Q - \sum W = \sum \Delta U$ on the totals line.

But what if... a ledger is only a layout until it is used? We try it on two paths between the same states: **Example 2**.



a: $2.0\,\mathrm{L}$, $1.5 \times 10^5\,\mathrm{Pa}$. b: $2.0\,\mathrm{L}$, $4.5 \times 10^5\,\mathrm{Pa}$. d: $6.0\,\mathrm{L}$, $4.5 \times 10^5\,\mathrm{Pa}$. c: $6.0\,\mathrm{L}$, $1.5 \times 10^5\,\mathrm{Pa}$. In process ab, $1800\,\mathrm{J}$ of heat is added; in bd, $2600\,\mathrm{J}$. Find (a) ΔU for ab, (b) ΔU for abd, (c) the total heat added along acd. (Any substance: the first law is not limited to ideal gases.)

Predict first

Path acd does less work than path abd. Will it need more heat or less heat than abd to reach the same state d?

Example 2

a: $2.0\,\mathrm{L}$, $1.5\times10^5\,\mathrm{Pa}$. b: $2.0\,\mathrm{L}$, $4.5\times10^5\,\mathrm{Pa}$. d: $6.0\,\mathrm{L}$, $4.5\times10^5\,\mathrm{Pa}$. c: $6.0\,\mathrm{L}$, $1.5\times10^5\,\mathrm{Pa}$. In process ab, $1800\,\mathrm{J}$ of heat is added; in bd, $2600\,\mathrm{J}$. Find (a) ΔU for ab, (b) ΔU for abd, (c) the total heat added along acd.

Step	Process	Q	W	ΔU
ab	isochoric	1800	0	1800
bd	isobaric	2600	1800	800
abd	total	4400	1800	2600

$W_{bd}=p\Delta V=(4.5\times10^5)(4.0\times10^{-3})=1800\,\mathrm{J}$. **Evaluate:** (a) $\Delta U_{ab}=1800\,\mathrm{J}$; (b) $\Delta U_{abd}=2600\,\mathrm{J}$.

But what if... path acd does less work than abd? Since ΔU is the same for both, the heat needed on acd follows from the first law alone.

Question: find the total heat added along acd (ac isobaric at $1.5 \times 10^5\,\mathrm{Pa}$, then cd isochoric), given $\Delta U_{abd} = 2600\,\mathrm{J}$.

Step	Process	Q	W	ΔU
ac	isobaric		600	
cd	isochoric		0	
acd	total	**3200**	600	2600

$W_{ac} = (1.5 \times 10^5)(4.0 \times 10^{-3}) = 600\,\mathrm{J}$; $\Delta U_{acd} = \Delta U_{abd}$, so $Q_{acd} = 2600 + 600$.

Evaluate: (c) $3200\,\mathrm{J}$. Less work means less heat is needed to reach the same state, as predicted ($3200 < 4400\,\mathrm{J}$). The empty cells could be filled only if we knew more about the substance. ✓

But what if... a path returns to its starting state, so that $\Delta U = 0$ for the whole trip? The first law then gives a simple result for a **cycle**.

Rule: the first law for a cycle

$$\Delta U = 0 \quad \Rightarrow \quad Q_{\text{net}} = W_{\text{net}} = \text{area enclosed}$$

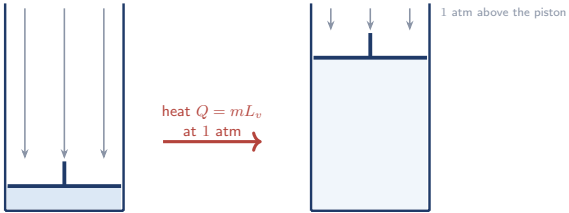
Clockwise: $W_{\text{net}} > 0$ and heat is converted to work overall. Counter-clockwise: $W_{\text{net}} < 0$ and work is converted to heat leaving the system.

Physical meaning

After a full cycle the system is unchanged, so nothing is stored. Whatever net work it does has been paid for by net heat taken in.

Common error

"In a cycle $Q = 0$ or $W = 0$."
Neither is zero. It is ΔU that is zero, so $Q_{\text{net}} = W_{\text{net}}$.



$3.0\,\mathrm{g}$ of water ($3.0\,\mathrm{cm}^3$) is boiled at $1.00\,\mathrm{atm}$ into steam of volume $5.01\,\mathrm{L}$ (a volume of $1671\,\mathrm{cm}^3$ per gram). The latent heat is $L_v = 2.256 \times 10^6\,\mathrm{J/kg}$. Find (a) the heat added, (b) the work done by the water, (c) the change in internal energy.

Predict first

The temperature stays at $100°\mathrm{C}$. Is most of the heat used to push back the atmosphere, or to change U?

Boiling water: the question

$3.0\,\mathrm{g}$ of water ($3.0\,\mathrm{cm}^3$) is boiled at $1.00\,\mathrm{atm}$ into steam of volume $5.01\,\mathrm{L}$ ($1671\,\mathrm{cm}^3$ per gram). The latent heat is $L_v = 2.256 \times 10^6\,\mathrm{J/kg}$. Find (a) the heat added, (b) the work done by the water, (c) the change in internal energy.

(a) $Q = mL_v = (3.0 \times 10^{-3})(2.256 \times 10^6) = 6.77 \times 10^3\,\mathrm{J}$.

(b) constant pressure, $p = 1.013 \times 10^5\,\mathrm{Pa}$:
$W = p\,(V_2 - V_1) = (1.013 \times 10^5)(5.01 \times 10^{-3} - 3 \times 10^{-6}) = 5.1 \times 10^2\,\mathrm{J}$.

(c) $\Delta U = Q - W = 6768 - 507 = 6.26 \times 10^3\,\mathrm{J}$.

Boiling water: where did the heat go?

	Energy (J)	Share of Q
heat added, Q	6770	100%
work against the atmosphere, W	507	7.5%
rise in internal energy, ΔU	6260	92.5%

Evaluate: only 7.5% of the heat is the work against the atmosphere; 92.5% goes into U. The molecular kinetic energy does not change (the temperature is constant), so the $6.26\,\mathrm{kJ}$ raises the *potential* energy of the molecules as they are pulled apart. Heat was added and the temperature did not rise. ✓

But what if... water turning to steam obeys the first law just as a gas does? The slips that cost marks are the same ones. We collect them.

Common mistakes and the correct habit

Mistake	Instead
Saying a gas "contains" heat or work.	It contains internal energy; heat and work are in transit.
Saying $Q = 0$ or $W = 0$ in a cycle.	$\Delta U = 0$, so $Q_{\text{net}} = W_{\text{net}}$.
Taking $\Delta U = Q$ in every process.	$\Delta U = Q - W$; $\Delta U = Q$ only at constant volume.

Take away: Q and W depend on the path; ΔU does not; over a cycle $\Delta U = 0$.

Contact A: what you can now do

- Name the system and apply the sign convention: $Q > 0$ into the system, $W > 0$ done by it.
- Find W from a pV diagram, with its sign, and show that W depends on the path.
- Apply $\Delta U = Q - W$ with an energy ledger, and treat a cycle: $\Delta U = 0$, $Q_{\text{net}} = W_{\text{net}}$.

Predict first

Before Wednesday, predict: an ideal gas is compressed to half its volume, once slowly in a water bath and once quickly in an insulated cylinder. In which case does the temperature rise? And is the heat needed per kelvin the same at constant volume as at constant pressure?

But what if... the first law is known, but not yet what it gives for each process of an ideal gas, nor why the heat needed per kelvin depends on the process? → **the four processes, heat capacities and adiabatic processes**.

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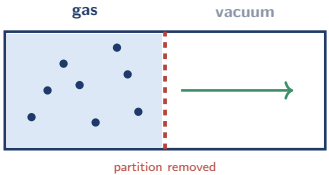
Contact B: Quantify, then Apply

What does the first law give for each process of an ideal gas, why does the heat needed per kelvin depend on the process, what is an adiabatic process, and how do we analyse a cycle?

Predict first

1. A gas expands from $2.0\,\mathrm{L}$ to $5.0\,\mathrm{L}$ at a constant $2.0 \times 10^5\,\mathrm{Pa}$. Find W, and say whether energy enters or leaves the gas as work.
   $W = p\Delta V = (2.0 \times 10^5)(3.0 \times 10^{-3}) = 600\,\mathrm{J}$; $W > 0$, so energy leaves the gas as work.
2. A gas absorbs $500\,\mathrm{J}$ of heat and does $200\,\mathrm{J}$ of work. Find ΔU.
   $\Delta U = Q - W = 500 - 200 = 300\,\mathrm{J}$.
3. A gas goes round a clockwise cycle that encloses $250\,\mathrm{J}$ on the pV diagram. Find ΔU, W_{net} and Q_{net}. $\Delta U = 0$; $W_{\text{net}} = +250\,\mathrm{J}$ (clockwise); so $Q_{\text{net}} = \Delta U + W_{\text{net}} = +250\,\mathrm{J}$.

But what if... no forces act between the molecules of an ideal gas, so there is no potential energy to raise? Then what does U depend on? The **internal energy of an ideal gas**.



In a **free expansion** the gas rushes into a vacuum: $W = 0$ (nothing to push), and with insulated walls $Q = 0$, so $\Delta U = 0$. Experiment: the temperature of a dilute gas does not change.

Rule: internal energy of an ideal gas

U depends on T only, not on p or V. For a monatomic ideal gas (no forces, so no potential energy; Lecture 10):

$$U = \tfrac{3}{2}nRT \qquad \Delta U = \tfrac{3}{2}nR\,\Delta T \quad \text{in every process.}$$

Question from earlier in Contact A

The gas goes from a $(2.0\,\mathrm{L},\ 4.0\times10^5\,\mathrm{Pa})$ to b $(6.0\,\mathrm{L},\ 1.0\times10^5\,\mathrm{Pa})$ by four paths, with works 1600, 1000, 879 and $400\,\mathrm{J}$. Are the heat inputs the same, and is anything the same on all four paths?

States a and b lie on the same isotherm ($p_aV_a = p_bV_b = 800\,\mathrm{J}$), so for an ideal gas $T_a = T_b$ and $\Delta U = 0$. The first law then gives $Q = W$ on every path:

Path	W (J)	ΔU (J)	Q (J)
(1)	1600	0	1600
(2)	1000	0	1000
(3)	879	0	879
(4)	400	0	400

Physical meaning

W and Q change from path to path, in step. $\Delta U = Q - W = 0$ does not. The gas must take in heat on every path ($Q = W > 0$, so "isothermal" does not mean "no heat flow"), because it does work and returns to the same internal energy.

But what if... a to b was a special case ($\Delta U = 0$)? We want W, ΔU, Q for each **process** of an ideal gas.

Rule: the first law for each process (monatomic gas, $\Delta U = \frac{3}{2}nR\Delta T$)

Process	Condition	W	ΔU	Q
isochoric	V constant	0	$\frac{3}{2}nR\Delta T$	$= \Delta U$
isobaric	p constant	$p\Delta V = nR\Delta T$	$\frac{3}{2}nR\Delta T$	$\frac{5}{2}nR\Delta T$
isothermal	T constant	$nRT\ln\frac{V_2}{V_1}$	0	$= W$
adiabatic	$Q = 0$	$= -\Delta U$	$\frac{3}{2}nR\Delta T$	0

Isobaric: $Q = \Delta U + W = \frac{3}{2}nR\Delta T + nR\Delta T$. In the adiabatic process the work comes out of U: the gas cools when it expands and warms when it is compressed (calculation later in Contact B).

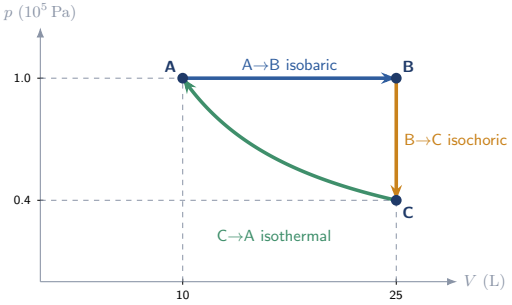
But what if... the table is only useful if we can tell which process a problem describes? We read the words and the diagram: **Toolkit 4**.

- **TOOLKIT 4 – Recognising the process**

1. READ THE WORDS: "rigid container" means V constant; "free piston under the atmosphere" or "heated in the open" means p constant; "slowly, in a thermal bath" means T constant; "insulated" or "very fast" means $Q \approx 0$.
2. READ THE DIAGRAM: vertical (isochoric), horizontal (isobaric), hyperbola $pV = \text{const}$ (isothermal), a curve steeper than the hyperbola (adiabatic).
3. WRITE THE CONSEQUENCE from the table: which of W, ΔU, Q is zero or known at once?
WATCH OUT: isothermal ($\Delta T = 0$) confused with adiabatic ($Q = 0$); using $\Delta U = 0$ for any process except an isotherm of an ideal gas.
CHECK: for an ideal gas, the sign of ΔU matches the sign of ΔT.

But what if... a problem chains several processes into one cycle? We combine the table and the ledger: **Example 3**.

Example 3: a three-process cycle



$0.40\,\mathrm{mol}$ of helium (a monatomic ideal gas) starts at A ($1.0 \times 10^5\,\mathrm{Pa}$, $10.0\,\mathrm{L}$). A→B: isobaric expansion to $25.0\,\mathrm{L}$. B→C: isochoric cooling until the temperature equals that at A. C→A: isothermal compression back to A. Find Q, W and ΔU for each leg.

Predict first

Sketch the cycle. Which legs absorb heat and which reject it? Is the net work positive?

Example 3

$0.40\,\text{mol}$ of helium (monatomic ideal gas) starts at A ($1.0 \times 10^5\,\text{Pa}$, $10.0\,\text{L}$). A→B: isobaric expansion to $25.0\,\text{L}$. B→C: isochoric cooling until the temperature equals that at A. C→A: isothermal compression back to A. Find Q, W and ΔU for each leg.

Set up: $p_A V_A = 1000\,\text{J}$, $T_A = \dfrac{1000}{(0.40)(8.314)} = 300.7\,\text{K}$; $T_B = \dfrac{(1.0 \times 10^5)(25.0 \times 10^{-3})}{(0.40)(8.314)} = 751.7\,\text{K}$; $T_C = T_A$ so $p_C = \dfrac{p_A V_A}{V_C} = 0.40 \times 10^5\,\text{Pa}$.

State	p (10^5 Pa)	V (L)	T (K)
A	1.00	10.0	300.7
B	1.00	25.0	751.7
C	0.40	25.0	300.7

But what if... we know all three states, but not yet the energy that crosses the boundary on each leg? Fill in the ledger, one row for each leg.

Question: Example 3: Q, W and ΔU for A→B (isobaric), B→C (isochoric) and C→A (isothermal) of $0.40\,\mathrm{mol}$ of helium.

Leg	Process	Q (J)	W (J)	ΔU (J)
A→B	isobaric	3750	1500	2250
B→C	isochoric	−2250	0	−2250
C→A	isothermal	−916	−916	0
cycle	total	584	584	0

AB: $W = p\Delta V = 1500$, $\Delta U = \frac{3}{2}(p_B V_B - p_A V_A) = \frac{3}{2}(2500 - 1000) = 2250$. BC: $\Delta U = \frac{3}{2}(1000 - 2500)$. CA: $W = 1000\ln(10/25) = -916$.

Evaluate: the cycle closes: $\sum \Delta U = 0$ and $\sum Q = \sum W = 584\,\mathrm{J}$, which is the enclosed area. Heat is absorbed only in AB. ✓

But what if... you had to do one such leg alone, say the isothermal compression C→A? Try it yourself.

Choose, then explain

An ideal gas is compressed to half its volume, once slowly in a large thermal bath and once very quickly in an insulated cylinder. Which statement is correct?

A. Isothermal: T rises. Adiabatic: T constant.
B. Isothermal: T constant and heat leaves the gas. Adiabatic: T rises and no heat flows.
C. Both: T constant.
D. Isothermal: no heat flows. Adiabatic: heat leaves the gas.
E. Both: T rises.

Predict first

Work is done on the gas in both. In which case can the energy leave as heat?

B. Isothermal: $\Delta U = 0$, so $Q = W < 0$. Adiabatic: $Q = 0$, so $\Delta U = -W > 0$ and T rises. Answer D swaps the two.

But what if... slips like this one cost marks in every contact? We collect them, with the habit that avoids each.

Poll

In the three-process cycle of Example 3, the isobaric heating A→B took $3750\,\mathrm{J}$ and the isochoric cooling B→C released $2250\,\mathrm{J}$. The temperature change of the $0.40\,\mathrm{mol}$ of helium was the same size in both ($451\,\mathrm{K}$). **Is the heat needed per kelvin the same at constant pressure and at constant volume?**

Predict first

Which process makes the gas do work as well as warm up? Is the ratio of the two heats a simple number?

The ratio is $3750/2250 = 5/3$. At constant pressure the heat must supply both the rise in U and the work of expansion. The heat per mole per kelvin: $\dfrac{3750}{(0.40)(451)} = 20.8$ against $\dfrac{2250}{(0.40)(451)} = 12.5\,\mathrm{J/(mol\,K)}$. These are the **molar heat capacities** C_p and C_V.

But what if... these need exact names? We define C_V, C_p, γ.

Defining the vocabulary of heat capacities

Quantity	What it is	What it measures	SI unit	Representation
Molar heat capacities C_V, C_p	the heat needed per mole for a small temperature change $\mathrm{d}T$: $\mathrm{d}Q = nC_V\,\mathrm{d}T$ at constant volume, $\mathrm{d}Q = nC_p\,\mathrm{d}T$ at constant pressure	how much heat a gas needs per kelvin in each case	$\mathrm{J/(mol\,K)}$	a table value
Ratio of heat capacities γ	the dimensionless ratio $\gamma = C_p/C_V$	how steeply the adiabat falls	none	$5/3$ or $7/5$

Interpret: the heat capacity answers *how much energy does this gas need per kelvin, and where does it go?*

But what if... C_p is larger than C_V? The first law says exactly by how much, and why.

Rule: molar heat capacities

$$Q_V = nC_V\Delta T, \qquad Q_p = nC_p\Delta T, \qquad \Delta U = nC_V\Delta T$$

$$C_p = C_V + R, \qquad \gamma = \frac{C_p}{C_V}, \qquad C_V = \frac{R}{\gamma - 1}$$

Q_V: constant volume. Q_p: constant pressure. ΔU: **every** process of an ideal gas.

Why $C_p = C_V + R$: at constant pressure, $Q = \Delta U + W$ gives $nC_p\Delta T = nC_V\Delta T + nR\Delta T$, since $W = p\Delta V = nR\Delta T$.

But what if... $C_p - C_V = R$ for every ideal gas, but what sets C_V itself? We compare with measurements.

Common error

$\Delta U = nC_V\Delta T$ holds for *every* process, not only at constant volume. $Q = nC_p\Delta T$ holds only at constant pressure.

Measured molar heat capacities

Type	Gas	C_V	C_p	γ
monatomic	He	12.47	20.78	1.67
	Ar	12.47	20.78	1.67
diatomic	H_2	20.42	28.74	1.41
	N_2	20.76	29.07	1.40
	O_2	20.85	29.17	1.40
polyatomic	CO_2	28.46	36.94	1.30

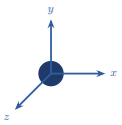
Measured values, Table 19.1 ($\mathrm{J/(mol\,K)}$).
$C_p - C_V \approx R = 8.31$ for all.

Physical meaning

Measured values agree with $\frac{3}{2}R$ and $\frac{5}{2}R$. Polyatomic molecules have extra terms and a larger C_V.

But what if... He and Ar match $\frac{3}{2}R$, but the diatomic gases lie near $\frac{5}{2}R$? Where does the extra energy go: **degrees of freedom**.

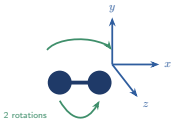
Degrees of freedom



monatomic (He, Ar)

3 translations: $f = 3$

$U = \frac{3}{2}nRT$, $C_V = \frac{3}{2}R$



diatomic (N_2, O_2)

3 translations + 2 rotations: $f = 5$

$U = \frac{5}{2}nRT$, $C_V = \frac{5}{2}R$

Definition: degrees of freedom

The number of velocity components needed to describe the motion of a molecule completely is its number of **degrees of freedom**, f.

At room temperature: monatomic $f = 3$ (v_x, v_y, v_z); diatomic $f = 5$ (three translations and two rotations; vibration is not excited).

But what if... we can count the ways a molecule moves, but not yet how much energy each way carries? That is the principle of **equipartition**.

Rule: equipartition

Each velocity component (linear or angular) has, on average, an associated kinetic energy per molecule of $\frac{1}{2}kT$. With f degrees of freedom:

$$U = \frac{f}{2}nRT, \quad C_V = \frac{f}{2}R, \quad \gamma = \frac{f+2}{f}$$

So monatomic $C_V = \frac{3}{2}R = 12.5$, $\gamma = \frac{5}{3}$; diatomic $C_V = \frac{5}{2}R = 20.8$, $\gamma = \frac{7}{5}$ (in $\mathrm{J/(mol\,K)}$). C_V is taken as constant: at low T rotations freeze out and at high T vibrations switch on (supplement).

But what if... we need every symbol and unit of these results in one place, ready for calculation? Two equation cards.

- **TOOLKIT 5 – Ideal-gas energy and heat capacities**

1. IDENTIFY THE GAS: count the degrees of freedom: $f = 3$ (monatomic), $f = 5$ (diatomic at room temperature); $C_V = \frac{f}{2}R$, $C_p = C_V + R$, $\gamma = C_p/C_V$.
2. ENERGY: $\Delta U = nC_V\Delta T$ for *every* process.
3. HEAT: $Q = nC_V\Delta T$ at constant V; $Q = nC_p\Delta T$ at constant p. In any other process, $Q = \Delta U + W$.
4. WORK: from Toolkit 2, or $W = nR\Delta T$ at constant pressure.

WATCH OUT: $Q = nC_V\Delta T$ outside a constant-volume process; $f = 5$ for a monatomic gas; °C differences are fine for ΔT but not for T.
CHECK: $Q_p > Q_V$ for the same ΔT, and $Q_p - Q_V = W = nR\Delta T$.

But what if... a toolkit is trusted only once it works on a real gas? **Example 4**: nitrogen at constant volume and at constant pressure.

Question

$1.5\,\mathrm{mol}$ of nitrogen gas ($\mathrm{N_2}$, treat as an ideal diatomic gas) is warmed from $300\,\mathrm{K}$ to $400\,\mathrm{K}$. Find Q, W and ΔU (a) at constant volume and (b) at constant pressure.

Predict first

In which case is more heat needed? By how much more, in terms of the work? Is ΔU the same in both cases?

Example 4

$1.5\,\mathrm{mol}$ of $\mathrm{N_2}$ (ideal diatomic gas) is warmed from $300\,\mathrm{K}$ to $400\,\mathrm{K}$. Find Q, W and ΔU (a) at constant volume and (b) at constant pressure.

$f = 5$: $C_V = \frac{5}{2}R = 20.79$, $C_p = \frac{7}{2}R = 29.10\,\mathrm{J/(mol\,K)}$; $\Delta T = 100\,\mathrm{K}$.
(a) $W = 0$; $\Delta U = Q = nC_V\Delta T = (1.5)(20.79)(100) = 3.12 \times 10^3\,\mathrm{J}$.
(b) $\Delta U = 3.12 \times 10^3\,\mathrm{J}$ (same ΔT); $Q = nC_p\Delta T = (1.5)(29.10)(100) = 4.36 \times 10^3\,\mathrm{J}$;
$W = nR\Delta T = 1.25 \times 10^3\,\mathrm{J}$.
Evaluate: $Q_p - Q_V = 1.25\,\mathrm{kJ} = W$, and $Q_p/Q_V = 1.40 = \gamma$. ✓

But what if... in both cases heat was allowed to flow in? What if no heat can flow and the gas still changes temperature: the **adiabatic process**.

- **Adiabatic** means $Q = 0$: no heat crosses the boundary. Either the walls are insulating, or the process is so fast that there is no time for heat to flow (but still slow enough for the gas to be in equilibrium at every step).
- The first law becomes $\Delta U = -W$: the work comes out of the internal energy.
- Expansion: $W > 0$ so U falls and the gas **cools**. Compression: the gas **warms**.
- Examples: a bicycle pump, a spray can, the compression in a diesel engine, and sound waves.

Sketch, then predict

From the same initial state, an ideal gas expands to twice its volume, once isothermally and once adiabatically. Sketch both on a pV diagram. Which ends at the lower pressure?

But what if... the gas cools when it expands adiabatically? We want to know by how much: the relation between p, V and T.

Deriving the adiabatic relation

1. **First law, $Q = 0$:** $\mathrm{d}U = -p\,\mathrm{d}V$, and for an ideal gas $\mathrm{d}U = nC_V\,\mathrm{d}T$. So $nC_V\,\mathrm{d}T = -p\,\mathrm{d}V$.
2. **Eliminate p** with $p = nRT/V$: $C_V\,\mathrm{d}T = -\dfrac{RT}{V}\,\mathrm{d}V$, that is $\dfrac{\mathrm{d}T}{T} = -\dfrac{R}{C_V}\dfrac{\mathrm{d}V}{V}$.
3. **Integrate;** with $R/C_V = \gamma - 1$ (from $C_p = C_V + R$): $\ln T + (\gamma - 1)\ln V = \text{constant}$.

Rule: adiabatic process of an ideal gas

$$TV^{\gamma-1} = \text{const}, \qquad pV^{\gamma} = \text{const}, \qquad W = nC_V(T_1 - T_2) = \frac{p_1V_1 - p_2V_2}{\gamma - 1}$$

Valid for an ideal gas with constant γ, in a quasi-static process with $Q = 0$. (Use $T = pV/nR$ to change between the first two.)

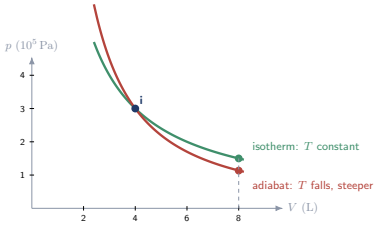
- **TOOLKIT 6 – Adiabatic scaling**

1. **VERIFY:** $Q = 0$ (insulated or fast), an ideal gas, a smooth quasi-static change.
2. **GET** γ from the degrees of freedom (monatomic $\frac{5}{3}$, diatomic $\frac{7}{5}$) or from the table.
3. **SCALE:** $\dfrac{T_2}{T_1} = \left(\dfrac{V_1}{V_2}\right)^{\gamma-1}$, $\dfrac{p_2}{p_1} = \left(\dfrac{V_1}{V_2}\right)^{\gamma}$; T in kelvin.
4. **WORK:** $W = nC_V(T_1 - T_2)$, or $W = -\Delta U$.
5. **SKETCH:** the adiabat is steeper than the isotherm through the same point.

WATCH OUT: using $pV =$ const (isothermal); using $^\circ\mathrm{C}$.
CHECK: expansion: $T_2 < T_1$ and $W > 0$; compression: $T_2 > T_1$ and $W < 0$.

But what if... the adiabat is said to be steeper than the isotherm through the same point? We compare them with numbers.

Isotherm against adiabat



A diatomic gas ($\gamma = 1.40$) expands from i ($4.0\,\mathrm{L}$, $3.0 \times 10^5\,\mathrm{Pa}$) to $8.0\,\mathrm{L}$.

	isotherm	adiabat
p_f (10^5 Pa)	1.50	1.14
T_f/T_i	1	0.758
W (J)	832	726

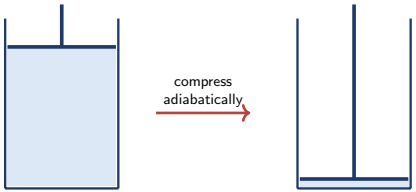
The adiabat ends lower: the gas cools as it does work, so its pressure falls faster. It also does less work, because the heat that the isothermal gas takes in is missing: $W_{\text{ad}} = -\Delta U$ comes out of U.

Check: $p_f = 3.0 \times 10^5 \times 2^{-1.4}$; $W_{\text{iso}} = 1200 \ln 2$.

But what if... adiabatic compression heats a gas this much in a real machine? The compression stroke of a **diesel engine**: Example 5.

Example 5: compression in a diesel engine

CORE



The compression ratio of a diesel engine is $15.0:1$: air in a cylinder is compressed adiabatically to $\frac{1}{15}$ of its initial volume. The initial pressure is $1.01\times10^5\,\mathrm{Pa}$, the temperature $27^\circ\mathrm{C}$ ($300\,\mathrm{K}$) and the volume $1.00\,\mathrm{L}$ ($C_V = 20.8\,\mathrm{J/(mol\,K)}$, $\gamma = 1.400$). Find (a) the final temperature, (b) the final pressure, (c) the work done by the gas.

Predict first

Will the final temperature be above or below $100^\circ\mathrm{C}$? Will the pressure rise by more or less than the factor 15?

Example 5

The compression ratio of a diesel engine is $15.0:1$: air in a cylinder is compressed adiabatically to $\frac{1}{15}$ of its initial volume. The initial pressure is $1.01\times10^5\,\text{Pa}$, the temperature 27°C ($300\,\text{K}$) and the volume $1.00\,\text{L}$ ($C_V = 20.8\,\text{J/(mol K)}$, $\gamma = 1.400$). Find (a) the final temperature, (b) the final pressure, (c) the work done by the gas.

Set up: $T_1 = 300\,\text{K}$, $p_1 = 1.01\times10^5\,\text{Pa}$, $V_1/V_2 = 15.0$;

$$n = \frac{p_1V_1}{RT_1} = \frac{(1.01\times10^5)(1.00\times10^{-3})}{(8.314)(300)} = 0.0405\,\text{mol};\ C_V = 20.8\,\text{J/(mol K)}.$$

(a) $$T_2 = T_1\left(\frac{V_1}{V_2}\right)^{\gamma-1} = 300\times15.0^{0.400} = 886\,\text{K} = \mathbf{613}^\circ\text{C}.$$

(b) $$p_2 = p_1\left(\frac{V_1}{V_2}\right)^{\gamma} = (1.01\times10^5\,\text{Pa})\times15.0^{1.400} = \mathbf{4.48\times10^6}\,\text{Pa} = 44\,\text{atm}.$$

But what if... the high temperature comes from the work done on the gas? Then how much work does the compression take?

Question: Example 5, part (c): the work done by the air ($0.0405\,\text{mol}$) in the adiabatic compression from $300\,\text{K}$ to $886\,\text{K}$.

(c) $W = nC_V(T_1 - T_2) = (0.0405)(20.8)(300 - 886) = \mathbf{-494}\,\text{J}$.

Evaluate: T_2 far above $100°\text{C}$ and p rises by 44, more than 15, because the temperature rises as well. $W < 0$ (compression). Check:
$T_2 = p_2V_2/(nR) = (4.48 \times 10^6)(6.67 \times 10^{-5})/(0.0405 \times 8.314) = 886\,\text{K}$, and
$(p_1V_1 - p_2V_2)/(\gamma - 1) = -494\,\text{J}$. ✓

But what if... the same relations apply to an expansion? Try one yourself.

Independent problem

$0.30\,\text{mol}$ of argon at $400\,\text{K}$ occupies $6.0\,\text{L}$. It expands adiabatically to $12.0\,\text{L}$. Find the initial pressure, the final temperature and pressure, and the work done by the gas.

Predict first

Sketch the isotherm and the adiabat. Is T_2 above or below $400\,\text{K}$? Is the work more or less than for an isothermal expansion?

Argon is monatomic: $\gamma = \frac{5}{3}$, $C_V = \frac{3}{2}R$. $p_1 = \dfrac{nRT_1}{V_1} = \dfrac{(0.30)(8.314)(400)}{6.0\times10^{-3}} = 1.66\times10^5\,\text{Pa}$.

$T_2 = 400\,(\frac{1}{2})^{2/3} = 252\,\text{K}$; $\quad p_2 = \frac{nRT_2}{V_2} = 5.24\times10^4\,\text{Pa}$ (check: $p_1(\frac{1}{2})^{5/3}$).

$W = nC_V(T_1 - T_2) = (0.30)(12.47)(148) = 554\,\text{J}$ (isothermal expansion: $nRT\ln 2 = 692\,\text{J}$).

Evaluate: cooler and a smaller work than the isothermal case, as predicted. ✓

But what if... isothermal and adiabatic work differ? Decide which compression needs more work.

Choose, then explain

An ideal gas is compressed to half its volume from the same initial state, once isothermally and once adiabatically. Which compression requires more work to be done *on* the gas?

A. the isothermal one B. the adiabatic one C. the same for both

Predict first

Sketch both paths on a pV diagram. Which path is higher, and what does the area under it represent?

B. Compression moves left. The adiabat is steeper, so its pressure is higher at every volume and the area under it is larger. For $\gamma = 1.40$: $|W_{\text{ad}}| = 2.5\,nRT_1(2^{0.4} - 1) = 0.80\,nRT_1$ against $|W_{\text{iso}}| = nRT_1 \ln 2 = 0.69\,nRT_1$.

But what if... each process is now under control one at a time? A real device combines them in a loop: **cycle analysis**, Toolkit 7.

- **TOOLKIT 7 – Cycle analysis**

1. STATES: find p, V and T at every corner, with $pV = nRT$ and, for an adiabatic leg, $pV^{\gamma} = \text{const}$.
2. LEGS: name the process on each leg (Toolkit 4).
3. LEDGER: for each leg find W (Toolkit 2), $\Delta U = nC_V\Delta T$ (Toolkit 5) and $Q = \Delta U + W$ (Toolkit 3).
4. CLOSE THE CYCLE: $\sum \Delta U = 0$ and $\sum Q = \sum W =$ the enclosed area.
5. SPLIT: add the positive Q (heat absorbed) and the negative Q (heat rejected) separately.

WATCH OUT: a corner state found from the wrong leg; the sign of W on a leg that goes to smaller V.
CHECK: the cycle closes; heat absorbed minus heat rejected equals the net work.

But what if... a method is worth having only if it is used? Where do cycles appear in engines, refrigerators and the atmosphere?

Connect: where the first law is used next

Lecture 12: engines

The cycle of the integrated problem delivers $331\,\mathrm{J}$ of work from $2000\,\mathrm{J}$ of heat. The efficiency of an engine and its limit follow from the second law.

Refrigerators

A counter-clockwise cycle moves heat from cold to hot with work: a refrigerator or heat pump.

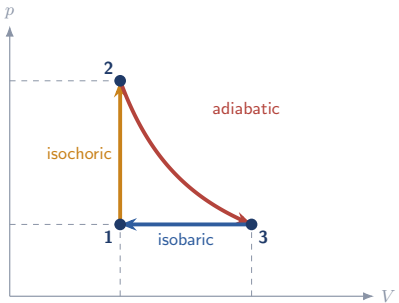
Atmosphere and engines

The adiabatic cooling of rising air sets weather and the lapse rate; adiabatic compression ignites fuel in a diesel engine.

Physical meaning

$\Delta U = Q - W$ is the conservation of energy of Lectures 4 and 5, with heat included.

But what if... every toolkit is needed at once, and nobody names the method? An **integrated problem**.



Application problem

$0.20\,\text{mol}$ of nitrogen (diatomic, ideal) starts at state 1: $1.00 \times 10^5\,\text{Pa}$, $4.0\,\text{L}$. The gas is heated at constant volume until its pressure is $3.00 \times 10^5\,\text{Pa}$ (state 2), then expands adiabatically until the pressure is again $1.00 \times 10^5\,\text{Pa}$ (state 3), then is compressed at constant pressure back to state 1.

(a) Find p, V and T at the three states.
(b) Find Q, W and ΔU for each leg.
(c) Find the net work and the net heat; compare.
(d) Which legs absorb heat?

Predict first

Sketch the cycle. Does it run clockwise? Predict the sign of W on each leg, and of the net work.

Application problem

$0.20\,\text{mol}$ of nitrogen (diatomic, ideal): state 1 is $1.00\times10^5\,\text{Pa}$, $4.0\,\text{L}$; heated at constant V to $3.00\times10^5\,\text{Pa}$ (state 2); adiabatic expansion to $1.00\times10^5\,\text{Pa}$ (state 3); constant-pressure compression back to state 1. (a) p, V, T at the states; (b) Q, W, ΔU for each leg; (c) net work and net heat; (d) which legs absorb heat?

Identify, set up: an ideal diatomic gas, a closed cycle: Toolkits 4, 5, 6 and 7. $\gamma = 1.40$, $C_V = 20.79$, $C_p = 29.10\,\text{J/(mol K)}$; $nR = 1.663\,\text{J/K}$.

State	p ($10^5\,\text{Pa}$)	V (L)	T (K)	How found
1	1.00	4.0	240.6	given; $T = pV/nR$
2	3.00	4.0	721.7	isochoric: $T_2 = 3T_1$
3	1.00	8.77	527.3	adiabat: $V_3 = V_2(p_2/p_3)^{1/\gamma} = 4.0\times3^{0.714}$

Check state 3: $T_3 = p_3V_3/nR = (1.00\times10^5)(8.77\times10^{-3})/1.663 = 527\,\text{K}$; also $T_3 = T_2(p_3/p_2)^{(\gamma-1)/\gamma} = 721.7\times3^{-0.2857} = 527\,\text{K}$. ✓

Question: Part (b): Q, W and ΔU for each leg of the nitrogen cycle (1→2 isochoric, 2→3 adiabatic, 3→1 isobaric).

Leg	Process	Q (J)	W (J)	ΔU (J)
1→2	isochoric	2000	0	2000
2→3	adiabatic	0	808	−808
3→1	isobaric	−1669	−477	−1192
cycle	total	331	331	0

12: $Q = nC_V(T_2 - T_1) = (0.20)(20.79)(481.1) = 2000$. 23: $W = nC_V(T_2 - T_3) = (0.20)(20.79)(194.4) = 808$. 31: $W = p\Delta V = (1.00 \times 10^5)(4.0 - 8.77) \times 10^{-3} = -477$; $Q = nC_p(T_1 - T_3) = (0.20)(29.10)(-286.7) = -1669$.

But what if... each leg has its Q, W and ΔU? We add the columns to answer parts (c) and (d): the net work, the net heat and which legs absorb heat.

Question: Parts (c) and (d): the net work and net heat of the nitrogen cycle, compared, and the legs that absorb heat.

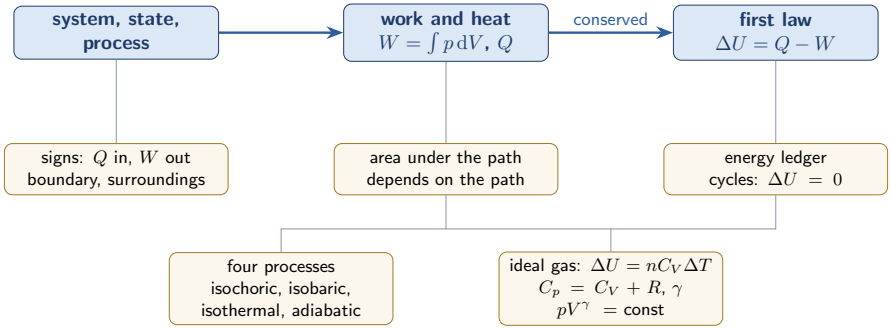
Evaluate: (a), (b) above; (c) $W_{\text{net}} = Q_{\text{net}} = 331\,\text{J}$, $\sum \Delta U = 0$: the cycle closes. (d) Only $1 \to 2$ absorbs heat ($2000\,\text{J}$); $3 \to 1$ rejects $1669\,\text{J}$. Clockwise, so $W_{\text{net}} > 0$, as predicted. Of the $2000\,\text{J}$ taken in, $331\,\text{J}$ (16.6%) becomes net work: this ratio is the *efficiency* of an engine, the subject of Lecture 12. ✓

But what if... the same adiabatic physics works outside the laboratory? A rising parcel of air is nearly adiabatic: an extension (unassessed).

Common mistakes and the correct habit

Mistake	Instead
Saying isothermal means no heat flow.	T constant: $Q = W$ for an ideal gas.
Using $\Delta U = 0$ for any process.	Only on an isotherm of an ideal gas.
Using $Q = nC_V\Delta T$ outside constant volume.	Use $\Delta U = nC_V\Delta T$; $Q = \Delta U + W$.
Using $pV =$ const for an adiabat.	$pV^\gamma =$ const.
Taking $f = 5$ for a monatomic gas.	$f = 3$ monatomic, 5 diatomic.
Using $^\circ$C in $TV^{\gamma-1}$.	Kelvin.
Finding a corner state from the wrong leg.	Name the process on each leg and follow its rule.

But what if... the cycle turned only 16.6% of the heat taken in into work? Could a better cycle turn all of it into work? That is the second law: Lecture 12.


system, state, process
work and heat
$W = \int p\,\mathrm{d}V$, Q
conserved
first law
$\Delta U = Q - W$
signs: Q in, W out
boundary, surroundings
area under the path
depends on the path
energy ledger
cycles: $\Delta U = 0$
four processes
isochoric, isobaric,
isothermal, adiabatic
ideal gas: $\Delta U = nC_V\Delta T$
$C_p = C_V + R$, γ
pV^γ = const

MASTER PROCEDURE – THE FIRST LAW

1. IDENTIFY The system, the substance (ideal gas? monatomic or diatomic?), and what is fixed.
2. SKETCH The path on a pV diagram, with arrows; label every state.
3. SIGNS Q in or out? W by or on the system? Units in joules and kelvin.
4. PREDICT The sign of W, Q, ΔU and ΔT for each leg.
5. SELECT THE MODEL Area or integral, first law, process rule, C_V and C_p, adiabat.
6. EXECUTE Fill in the ledger, one row for each leg, equation first.
7. EVALUATE Units, signs, ΔU the same on every path, the cycle closes, limiting cases.

SKETCH $\rightarrow$ PREDICT $\rightarrow$ CALCULATE $\rightarrow$ CHECK.

Inside the course framework

IDENTIFY $\leftrightarrow$ steps 1–2
SET UP $\leftrightarrow$ steps 3–5
EXECUTE $\leftrightarrow$ step 6
EVALUATE $\leftrightarrow$ step 7

Physical meaning

Next lecture: the cycle of the integrated problem delivers net work $331\,\mathrm{J}$ from $2000\,\mathrm{J}$ of heat. Why can the ratio never reach one? That is the second law, heat engines and entropy.

• THE SEVEN TOOLKITS

1	**System and sign convention**	$Q > 0$ in; $W > 0$ by the system; joules
2	**Work from a pV diagram**	area; sign by direction; cycles
3	**The energy ledger**	$\Delta U = Q - W$ row by row; state function
4	**Recognising the process**	isochoric, isobaric, isothermal, adiabatic
5	**Ideal-gas energy and heat capacities**	$\Delta U = nC_V\Delta T$; $C_p = C_V + R$; f
6	**Adiabatic scaling**	$TV^{\gamma-1}$, pV^{γ} constant; steeper than isotherm
7	**Cycle analysis**	states, ledger per leg, closure

The Master Procedure sits above these. Earlier toolkits (units, energy, signs, the ideal-gas equation, the ratio method, $Q = mc\Delta T$) are reused, not renamed.

Key ideas

- Q and W depend on the path; ΔU does not.
- $W > 0$ on expansion; $Q > 0$ for heat in.
- For a cycle, $\Delta U = 0$ and $Q_{\text{net}} = W_{\text{net}}$.
- For an ideal gas, $\Delta U = nC_V \Delta T$ in every process.
- Isothermal: T constant; adiabatic: $Q = 0$.
- The adiabat is steeper than the isotherm.

Where to look

Definitions: the vocabulary slides of each contact; the equation-anatomy slides are in the Reference deck.

Common mistakes: the last slides of Contacts A and B.

Connections: the Connect boxes and slides.

Summary: core equations

Idea	Equation	Conditions and units
Work by a gas	$W = \int_{V_1}^{V_2} p\,\mathrm{d}V$	area; $W = p\Delta V$ at constant p
Isothermal work	$W = nRT\ln(V_2/V_1)$	ideal gas; J
First law	$\Delta U = Q - W$	$Q > 0$ in, $W > 0$ by the system
Cycle	$Q_{\text{net}} = W_{\text{net}}$	$\Delta U = 0$
Ideal-gas energy	$U = \frac{f}{2}nRT, \quad \Delta U = nC_V\Delta T$	any process
Heat capacities	$Q_V = nC_V\Delta T, \quad Q_p = nC_p\Delta T, \quad C_p = C_V + R$	J/(mol K)
Adiabatic	$pV^\gamma = \text{const}, \quad TV^{\gamma-1} = \text{const}$	$Q = 0$, ideal gas
Adiabatic work	$W = nC_V(T_1 - T_2)$	J

Checklist: can I . . .

CORE

- ☐ name the system and classify it as open, closed or isolated?
- ☐ state the sign convention for Q and W?
- ☐ derive and use $W = \int p\,\mathrm{d}V$?
- ☐ find the work from a pV diagram, with its sign?
- ☐ explain why W depends on the path?
- ☐ describe internal energy as a state function?
- ☐ apply $\Delta U = Q - W$ and build a ledger?
- ☐ treat a cycle (net work equals net heat)?
- ☐ recognise the four processes and give W, Q, ΔU?
- ☐ use the internal energy of an ideal gas?
- ☐ use C_V, $C_p = C_V + R$ and γ?
- ☐ relate C_V to the degrees of freedom?
- ☐ use $pV^\gamma = \text{const}$ and $TV^{\gamma-1} = \text{const}$?
- ☐ compare an adiabat with an isotherm?
- ☐ analyse a cycle for net work and net heat?
- ☐ select the model in an unfamiliar problem?

After this week

Tutorial sheet: signs, work from areas, ledger and cycle problems, the four processes, boiling water, heat capacities, adiabatic problems and a combined problem where the method is not given. Supplementary notes: Joule's experiment, the isothermal and adiabatic derivations, C_V against temperature, polyatomic gases, the sign convention $\Delta U = Q + W$, enthalpy, and a preview of Lecture 12. The Reference deck holds the equation-anatomy slides and the practice items cut from the live lecture.

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